NAME: _____ Geometry (FL B.E.S.T.)

1.4 Pairs of Angles

Complementary, supplementary, vertical, and linear-pair angles are the building blocks behind almost every later angle proof, including parallel-line theorems in Chapter 2. The key upgrade here is recognizing that these relationships are theorems and postulates you can name and chain together — for instance, two angles vertical to the same angle must be congruent to each other, not just each congruent to the middle one. Expect algebra with the unknown inside the angle expressions, and at least one problem where you must find all four angles at an intersection from a single given value.

Practice Set — All Questions

15 questions. Work them on your own paper, then check the key below.

1. Complement of $42°$?

2. Supplement of $125°$?

3. $2x$ and $x + 30$ are vertical. Find x.

4. $3x$ and $x + 40$ form a linear pair. Find x.

5. $x + 20$ and $2x + 10$ complementary. Find x.

6. Vertical to $68°$?

7. $7x$ and $2x + 9$ supplementary. Find x.

8. Complement of $15°$?

9. Supplement of $90°$?

10. $5x - 10$ and $3x + 30$ linear pair. Find x.

11. $4x + 8$ and $6x - 12$ vertical. Find x.

12. $4x$ and $5x + 9$ complementary. Find x.

13. Angles that share a vertex and side but do not overlap are called what?

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14. Two lines intersect. One angle is $(3x + 10)^\circ$ and its vertical angle is $(5x - 30)^\circ$. Find x, then find the measure of all four angles formed.

15. Two angles are supplementary. The larger angle is 10° less than 4 times the smaller. Find both measures.

✓ Answer Key

1. 48°
2. 55°
3. $x = 30$
4. $x = 35$
5. $x = 20$
6. 68°
7. $x = 19$
8. 75°
9. 90°
10. $x = 20$
11. $x = 10$
12. $x = 9$
13. Adjacent
14. $x = 20$; angles are 70°, 110°, 70°, 110°
15. 38° and 142°